

The Miracle on the Han

1 Background

Korea’s Heavy and Chemical Industrialisation. Between 1962 and 1979 South Korea pursued a state-directed export industrialisation strategy under a series of five-year plans. The pivot from light to heavy industry began in earnest with President Park Chung-hee’s *Heavy and Chemical Industry* (HCI) declaration of 1973, which committed public investment to six strategic sectors: steel, non-ferrous metals, machinery, shipbuilding, electronics, and chemicals. Gross fixed capital formation in manufacturing rose from 5 % to over 10 % of GDP within a decade. The structural shift is visible in the national accounts: manufacturing’s share of GDP rose from around 15 % in 1965 to nearly 30 % by 1985, primarily at the expense of agriculture (Figure 1). This transformation was among the fastest recorded anywhere in the post-war period.

Industrial parks as the spatial delivery mechanism. Korea’s industrial policy was implemented primarily through purpose-built *industrial complexes* () — planned estates with shared infrastructure, tax incentives, and preferential credit for designated firms. The legal framework traces to the Industrial Sites and Development Act of 1964, and the first large-scale complex at Guro (Seoul) opened in 1967. During the HCI push, new parks were established in Changwon (machinery, 1974), Pohang (steel, 1968), Onsan (non-ferrous metals, 1974), and a cluster of fabricated-metals and electronics parks along the Gyeongbu motorway corridor. Figure 4 shows the spatial expansion from 1965 to 1990: the parks were concentrated in the southeastern coastal corridor and the Seoul–Busan axis, exploiting proximity to deepwater ports and the newly built highway network. By 1985 eight of the twenty parks recorded in our database had been established within the fabricated metals sector (industry code 28), the focus of our SDID analysis.

Roads and market integration. Road construction was both a complement and a precondition for the industrial parks strategy. The Gyeongbu Expressway (Seoul–Busan, 428 km) was completed in 1970, followed by five additional motorways by 1981.

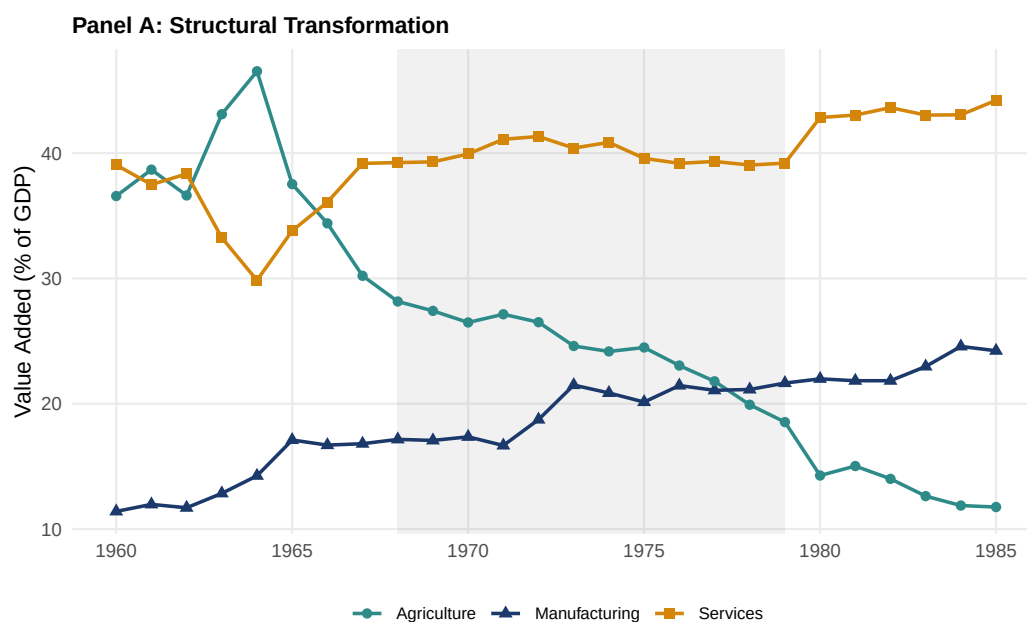
Our data show that highway coverage roughly doubled between 1970 and 1980 (Figure 4). For rural townships, these roads transformed effective distances to urban markets, enabling both the outward migration of farm labour and the inward penetration of manufactured inputs and consumer goods.

Geographic concentration of manufacturing. The industrial parks strategy produced a highly concentrated spatial pattern of manufacturing activity. Figure 3 shows log production in the fabricated metals sector at the county level in 1968 and 1985, with active industrial parks overlaid as orange dots. In 1968, before the HCI drive, production was dispersed with a modest concentration around Seoul. By 1985, it had shifted decisively towards the southeastern coastal corridor — Changwon, Ulsan, Pohang, and the Gyeongbu axis — where the major HCI parks were located. The co-location of high-production counties and park locations is striking and motivates the county-level SDID analysis in Section 3.1.

Transport infrastructure expansion. The parks were complemented by rapid road construction. Figure 5 shows paved road and motorway coverage at four points in time. The Gyeongbu Expressway (Seoul–Busan) opened in 1970, and by 1975 a second arc of motorways had connected Daejeon, Daegu, and the port of Masan. Paved local road coverage expanded in parallel, with density roughly tripling between 1967 and 1980. This infrastructure build-out reduced travel times to urban markets for rural townships throughout the country, forming the *market access* channel that is the focus of the second part of our empirical analysis.

Structural transformation and rural change. The industrial and infrastructure push was accompanied by rapid urbanisation and agricultural contraction. Korea’s urban population share rose from approximately 28 % in 1960 to 57 % by 1980 and 74 % by 1990. The farm household population fell from a peak of roughly 14 million in the early 1960s to under 6 million by 1990 as workers relocated to industrial centres. Internal migration flows — dominated by moves from rural provinces to Seoul, Busan, and the industrial corridor — averaged 4–8 million persons per year through the 1970s and 1980s. Against this backdrop, understanding whether the gains from industrialisation reached rural areas — and through what channels — is the central empirical question of this paper.

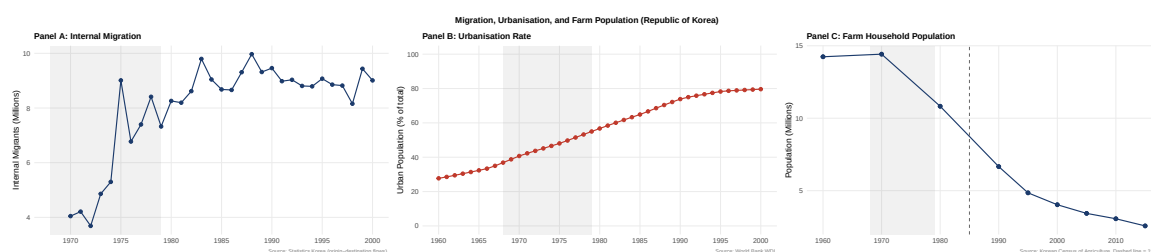
Figure 1: Structural Transformation and Rural Change



Source: World Bank WDI.

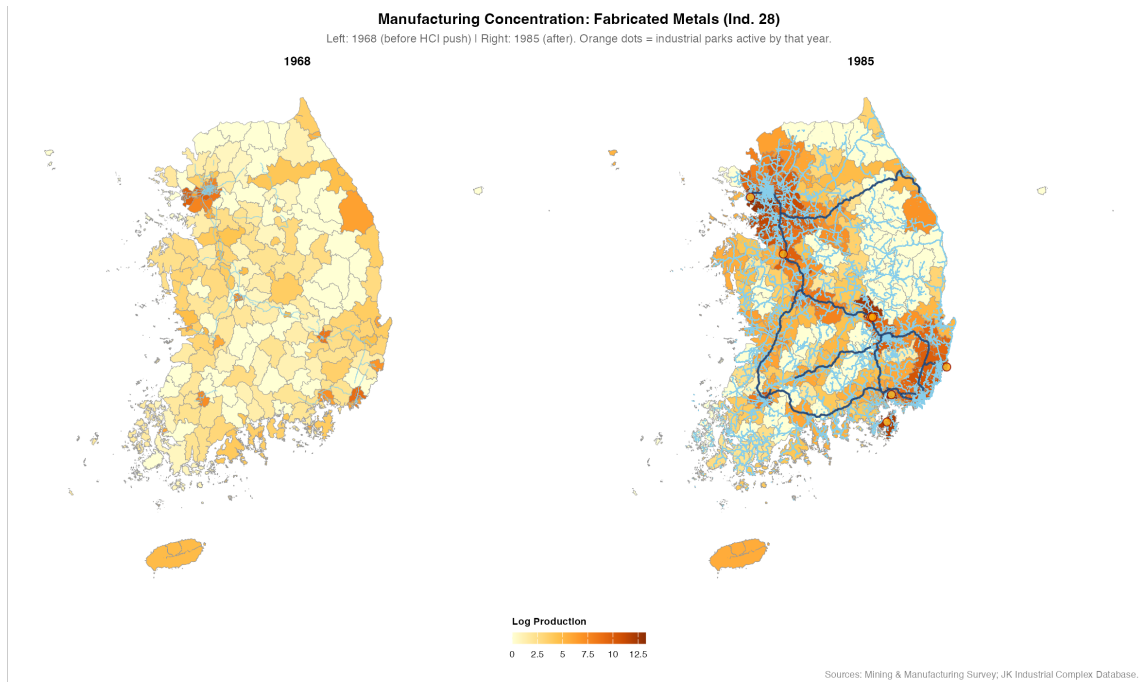
Notes: Share of GDP by sector (agriculture, manufacturing, services), 1960–1985. Grey band = HCI period (1968–1979). Source: World Bank WDI.

Figure 2: Internal Migration, Urbanisation, and Farm Population



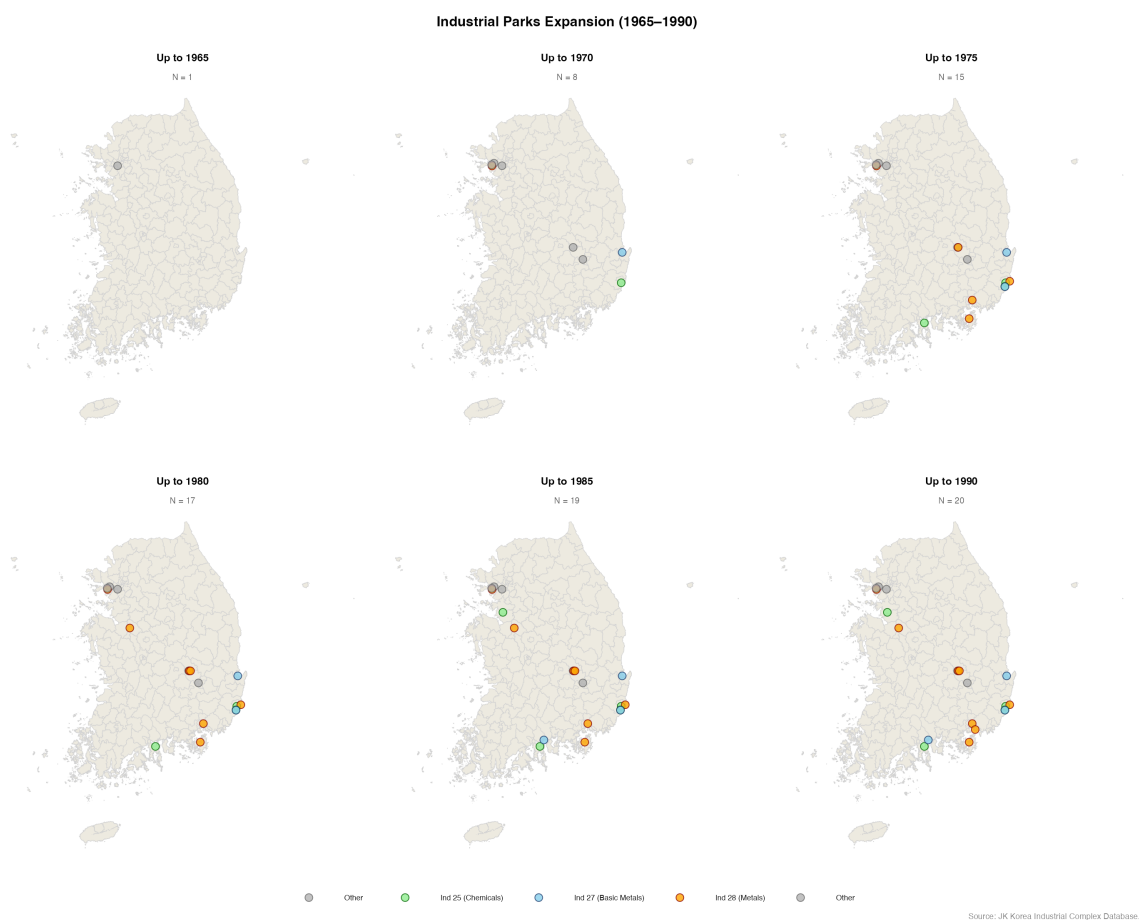
Notes: Panel A: Annual internal migrants (millions). Panel B: Urban population share (%). Panel C: Farm household population (millions). Grey band = HCI period (1968–1979). Sources: Statistics Korea; World Bank WDI; Korean Census of Agriculture. Dashed line in Panel C: 1985.

Figure 3: Manufacturing Concentration and Industrial Parks, 1968 vs. 1985



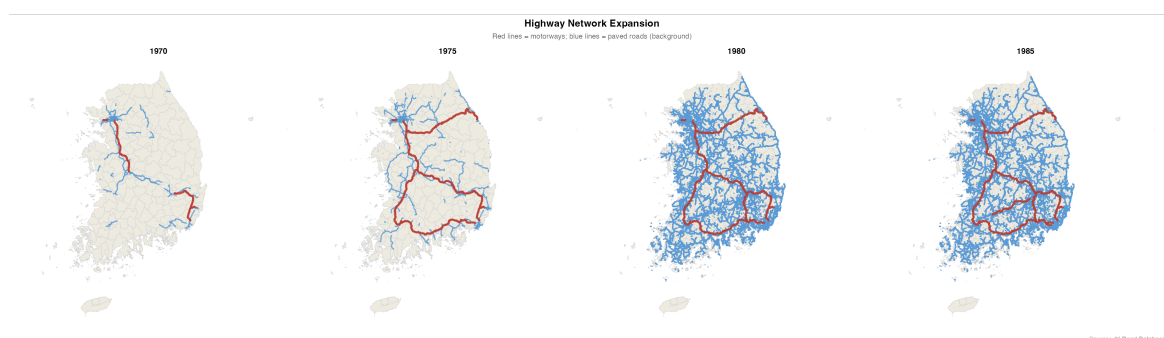
Notes: Log total production in the fabricated metals sector (Ind. 28) by county. Colour scale: yellow–brown = YlOrBr, fixed across years. Orange dots = industrial parks with groundbreaking $\leq$ the panel year. Blue lines = paved roads; navy lines = motorways. Source: Mining & Manufacturing Survey (harmonized)

Figure 4: Industrial Park Expansion by Sector, 1965–1990



Notes: Location of industrial parks at six points in time, coloured by target industry. Orange = fabricated metals (Ind. 28); green = chemicals (Ind. 25); blue = basic metals (Ind. 27).

Figure 5: Transport Infrastructure Expansion, 1970–1985



Notes: Road network at four snapshots. Red lines = motorways (expressways) built up to each year; blue lines = paved roads. County boundaries shown in grey.

2 Empirical Strategy

Our analysis proceeds in two stages that match the two channels through which the HCI policy affected rural Korea. The first stage asks whether industrial parks actually raised manufacturing activity in the counties that hosted them — a necessary precondition for any rural spillover. The second stage estimates how road-driven improvements in *market access* propagated these gains to surrounding rural townships.

2.1 Local Manufacturing Effects: Synthetic DiD

To identify the direct effect of hosting an industrial park on county-level manufacturing, we use the Synthetic Difference-in-Differences estimator of ?. SDID simultaneously reweights control units (to match each treated county’s pre-treatment outcome level) and pre-treatment time periods (to downweight periods in which treated and control trends diverge), providing better pre-treatment fit than either pure synthetic control or standard two-way fixed effects. The estimator reduces to a standard DiD when all pre-treatment weights are equal, and to a synthetic control when all period weights are equal; in our setting the double reweighting exploits both cross-sectional and time-series structure.

The sample is the harmonised Mining and Manufacturing Survey (MMS) county-year panel, 1968–1990, covering 183 counties. The outcome is county-level production (and employment) in fabricated metals (industry code 28) — the dominant HCI machinery sector. Treatment is the groundbreaking of an industrial park specialised in that sector. We identify eight groundbreaking cohorts (1969, 1971, 1972, 1973, 1974, 1977, 1979, 1986), each with a single treated county and 175 donor counties. Cohort-level SDID estimates are aggregated to a weighted event study using the number of treated counties per cohort as weights. Standard errors are constructed from a 100-replicate county cluster bootstrap that mirrors the Stata `sdid` package implementation. The key identifying assumption is that, absent the park, each treated county’s trajectory would have continued as reconstructed by the SDID weights; the flat pre-treatment event-study coefficients reported in Section 3.1 provide direct evidence for this assumption.

2.2 Rural Spillovers: Market Access Framework

Korea’s HCI drive simultaneously expanded industrial parks and the paved road network. To isolate the rural spillovers of improved *market access* from the direct effects of industrial park proximity, we adopt the market access framework of ? in a first-

differences specification.

Market access. Following ? and ?, market access for rural township i at time t is the trade-cost-weighted sum of urban market sizes:

$$\text{MA}_{i,t} = \sum_{c \in \mathcal{C}} \tau_{i,c,t}^{-\theta} P_{c,t} \quad (1)$$

where $\tau_{i,c,t}$ is travel time to city c via the paved road network, $\theta = 1$ is the trade elasticity, $P_{c,t}$ is city population, and $\mathcal{C}$ denotes 28 urban destinations. We estimate the first-difference equation:

$$\Delta Y_i = \beta \Delta \log(\text{MA}_i) + \mathbf{X}'_{i,1970} \Gamma + \Delta \varepsilon_i \quad (2)$$

where ΔY_i is the outcome change between a baseline period (1970 or 1975) and 1990, and $\mathbf{X}_{i,1970}$ includes log road efficiency, log population density, reservoir area, caloric suitability, total area, distance to water, slope, ruggedness, elevation, and log travel cost to the nearest industrial park. Standard errors are clustered at the county level.

Identification. Road expansion during 1974–1987 was not random. To address endogeneity, we fix city populations at 1970 levels and exclude the J nearest destination cities, removing the mechanical correlation between local road improvements and the MA measure:

$$\Delta \log(\widetilde{\text{MA}}_i^J) = \ln \left(\sum_{c: \text{rank}(i,c) \geq J} \tau_{i,c,1987}^{-\theta} P_{c,1970} \right) - \ln \left(\sum_{c \in \mathcal{C}} \tau_{i,c,1974}^{-\theta} P_{c,1970} \right) \quad (3)$$

where $J \in \{3, 5, 8, 10\}$. Our preferred specification sets $J = 5$; stability of estimates across thresholds is the primary robustness check. All specifications additionally control for local paved road density within a 30 km buffer. We examine five outcome domains: (i) agricultural machinery adoption, (ii) agricultural productivity, (iii) labour intensity, (iv) rural population and migration by farm status and age cohort, and (v) service sector growth. The sample covers 1,307–1,390 townships depending on outcome availability.

3 Results

3.1 Direct Effects of Industrial Parks on Manufacturing Output and Employment

Section 2.1 describes the SDID design. We report the empirical results here.

Aggregate event study. Figure 6 presents the aggregate event-study for production (left panel) and employment (right panel). Three features stand out. First, pre-treatment coefficients are uniformly small and statistically indistinguishable from zero across all event-time horizons $k \in \{-6, \dots, -1\}$, supporting the parallel-trends assumption that underlies SDID. Second, both outcomes exhibit a sharp and persistent post-treatment increase beginning at $k = 0$ (the year of groundbreaking). Third, the production and employment effects co-move, consistent with genuine output expansion rather than capital-labour substitution.

Figure 6: Synthetic DiD Event Study: Direct Effects of Industrial Parks on Manufacturing (Ind. 28)



Notes: Aggregate SDID event study for fabricated metals (industry 28). Left: total production (MMS survey units). Right: employment. Each cohort SDID uses 175 donor counties; cohorts are aggregated weighted by treated-county count. Shaded band = 95% confidence interval from 100-replicate county cluster bootstrap. Normalised so that $k = -1$ equals zero.

Cohort-level ATT estimates. Table 1 reports cohort-specific ATT estimates. The largest effects are concentrated in the early HCI cohorts: parks established in 1972 and 1973 raised production by approximately 356,000 and 399,000 MMS units respectively, and employment by 34,000 and 55,000 workers relative to the synthetic control. These cohorts correspond to the founding of Changwon (machinery) and the Onsan–Ulsan complex — the largest industrial estates built during the HCI period. Later cohorts (1977, 1986) show substantially smaller effects, consistent with diminishing returns as the best industrial sites were occupied and as the HCI policy was wound down after 1979. The 1979 cohort, established at the onset of the second oil shock, shows a near-zero effect for both outcomes, suggesting that macro headwinds substantially attenuated the local impact for latecomers. Taken together, the SDID estimates confirm that Korea’s industrial parks delivered large, targeted manufacturing gains in the treated counties — the “first stage” of any rural spillover analysis. The subsequent sec-

tions examine whether these gains propagated to surrounding rural townships through improved market access.

Table 1: SDID Cohort-Level ATT Estimates: Fabricated Metals Production and Employment (Ind. 28)

Cohort	Production (ATT)	Employment (ATT)
1969	<i>(estimated; cohort map available)</i>	
1971	197,819	20,353
1972	355,929	33,708
1973	399,398	55,222
1974	78,531	10,625
1977	21,424	336
1979	-1,466	-647
1986	13,441	1,985

Notes: SDID point estimates for each treatment cohort. Outcome: total manufacturing output (production) and employment in industry 28 (fabricated metals), from the harmonised Mining and Manufacturing Survey. Each cohort uses 175 never-treated donor counties. ATT values are in MMS survey units (millions of Korean won at current prices for production; number of workers for employment). The 1969 cohort estimate is available in supplementary figures. $k = 0$ corresponds to the groundbreaking year of the industrial park.

3.2 Agricultural Modernisation

Machinery adoption. Table 2 reports OLS estimates for machinery adoption. Column (1) uses the raw MA measure; columns (2)–(6) fix population and progressively exclude nearby cities; columns (7)–(8) use total tillers and the HH adoption share as outcomes. The estimates are large, positive, and stable across all MA specifications. The preferred estimate (col. 4, $J = 5$) implies 272 additional households owning tillers per unit of $\Delta \log(\widetilde{\text{MA}}_i^5)$ (s.e. 68, $t = 3.98$); the tiller adoption share rises by 0.19 percentage points (s.e. 0.05, $t = 3.86$). The R^2 of 0.45 indicates that geography explains much of the cross-sectional variation in adoption, with market access delivering an economically large additional push. The insensitivity to J across columns rules out the concern that results are driven by local road improvements near any particular subset of cities.

Table 2: Agricultural Machinery Adoption, 1970–1990

Table 3:

	(1) HH tillers	(2) HH tillers	(3) HH tillers	(4) HH tillers [*]	(5) HH tillers	(6) HH tillers	(7) N tillers	(8) HH share
log(D(74-87) MA)	131.470** (64.167)	200.824*** (63.243)						
log(D MA excl. 3)			259.568*** (65.330)					
log(D MA excl. 5) [preferred]				271.607*** (68.271)			270.252*** (70.665)	0.188*** (0.049)
log(D MA excl. 8)					292.782*** (81.406)			
log(D MA excl. 10)						287.036*** (86.304)		
Observations	1307	1307	1307	1307	1307	1307	1307	1298
R ²	0.415	0.453	0.457	0.457	0.455	0.454	0.452	0.379
Controls	Geography	Geography + IP dist.	Geography + IP dist.	Geography + IP dist.	Geography + IP dist.	Geography + IP dist.	Geography + IP dist.	Geography + IP dist.

Notes: OLS estimates of equation (2). Dep. var.: change in households owning agricultural tillers (cols. 1–6), change in total number of tillers (7), and change in the HH tiller adoption share (8). All columns include geography controls (crop suitability, ruggedness, slope, elevation, distance to water, reservoir area, log population, log area, log efficient road). Cols. (2)–(8) additionally control for log distance to the nearest industrial park. [*] marks the preferred specification ($J = 5$). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; standard errors clustered by county.

Productivity and labour intensity. Table 4 turns to agricultural productivity and labour use. Market access growth raises value added per worker and agricultural productivity by approximately 0.70 log points (s.e. 0.30, $t = 2.33$) while reducing labour intensity by 0.61 log points (s.e. 0.29). These estimates form a coherent narrative alongside the machinery results: improved road connectivity induces mechanisation, which raises output per worker while reducing the labour applied per unit of land. Township agriculture becomes simultaneously more capital-intensive, more productive, and less labour-dependent — the canonical pattern of agricultural modernisation. The results are stable across MA exclusion thresholds (see cols. 1–5 for value added).

Table 4: Agricultural Productivity and Labour Intensity, 1970–1990

Table 5:

	(1) Value added	(2) Value added	(3) Value added [*]	(4) Value added	(5) Value added	(6) Ag. prod.	(7) Labour int.
log(D(74-87) MA)	0.397 (0.242)						
log(D MA excl. 3)		0.535* (0.274)					
log(D MA excl. 5) [preferred]			0.703** (0.302)			0.703** (0.302)	-0.610** (0.293)
log(D MA excl. 8)				0.780** (0.356)			
log(D MA excl. 10)					0.826** (0.373)		
Observations	1298	1298	1298	1298	1298	1298	1289
R ²	0.101	0.105	0.111	0.110	0.110	0.111	0.173

Notes: OLS estimates of equation (2). Outcomes are log changes, 1970–1990. Controls as in Table 2. [*] marks the preferred specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; SE clustered by county.

3.3 Rural Population and Migration

Farm and non-farm population. To capture the role of industrial parks as competing labour-demand destinations, we augment equation (2) with an interaction between $\Delta \log(\widetilde{\text{MA}}_i^J)$ and log distance to the nearest industrial park:

$$\Delta Y_i = \beta_1 \Delta \log(\widetilde{\text{MA}}_i^J) + \beta_2 \Delta \log(\widetilde{\text{MA}}_i^J) \times \ln \text{IP dist}_i + \beta_3 \ln \text{IP dist}_i + \mathbf{X}'_{i,1970} \Gamma + \Delta \varepsilon_i \quad (4)$$

Table 6 reports OLS estimates for non-farm (cols. 1–3), farm (cols. 4–6), and total population (col. 7) between 1975 and 1990. The main effect $\hat{\beta}_1$ is negative and significant for non-farm population, but the positive interaction coefficient $\hat{\beta}_2$ implies that this effect reverses at sufficient distance from industrial parks. The total marginal effect $\hat{\beta}_1 + \hat{\beta}_2 \cdot \ln \text{IP dist}_i$ is plotted in Figure 7: townships in the lowest quartile of IP distance experience strong labour outmigration (MA accelerates reallocation to nearby industrial jobs), while remote townships exhibit modest non-farm retention (improved access raises local demand). Farm population effects are more muted; service sector growth (Section 3.4) mediates the non-farm retention channel in remote areas.

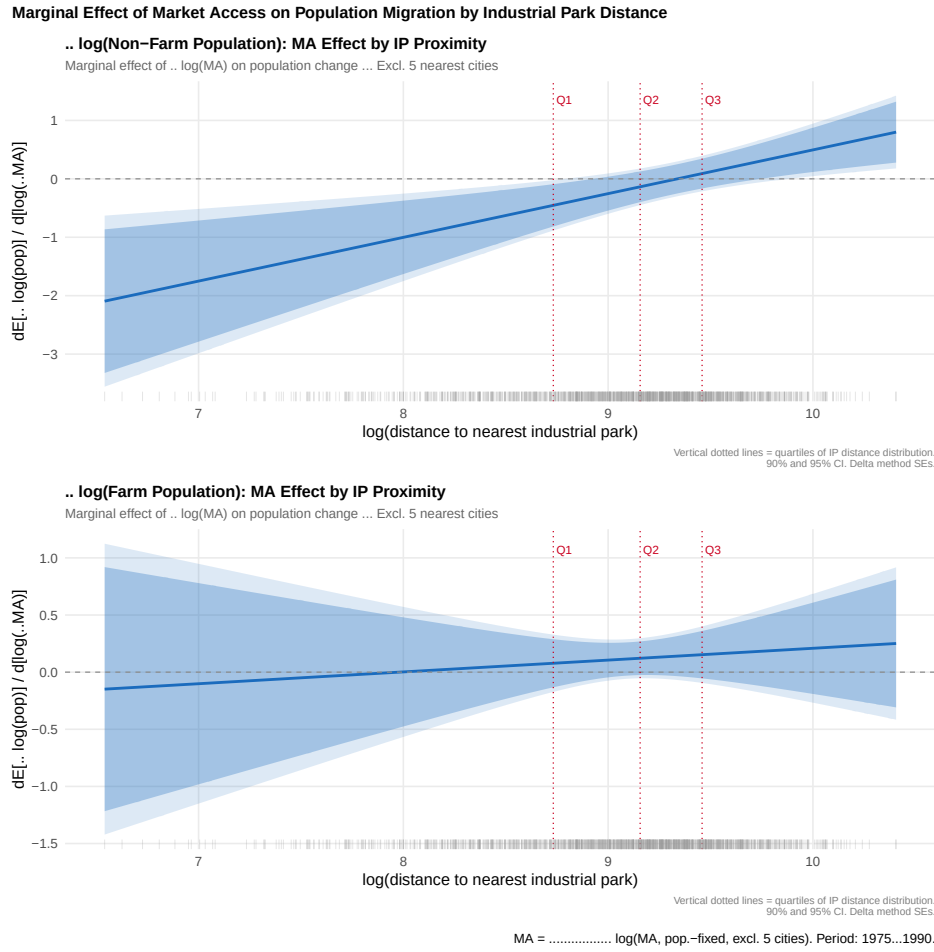
Table 6: Population Migration, 1975–1990: Market Access $\times$ Industrial Park Distance

Table 7:

	(1) Non-farm	(2) Non-farm	(3) Non-farm	(4) Farm	(5) Farm	(6) Farm	(7) Total
log(D MA excl. 3)	-6.051** (2.409)			-1.229 (2.154)			
log(D MA excl. 5) [preferred]		-6.991*** (2.445)			-0.825 (2.290)		
log(D MA excl. 10)			-8.606*** (2.887)			0.089 (2.117)	-5.227** (2.029)
log(D MA excl. 3) x log(dist. IP)	0.648** (0.257)			0.147 (0.237)			
log(D MA excl. 5) x log(dist. IP)		0.749*** (0.261)			0.103 (0.251)		
log(D MA excl. 10) x log(dist. IP)			0.912*** (0.306)			0.015 (0.238)	0.576*** (0.216)
Observations	1306	1306	1306	1306	1306	1306	1306
R ²	0.225	0.228	0.229	0.026	0.025	0.026	0.371

Notes: OLS estimates of equation (4). Dep. var.: log change in non-farm population (cols. 1–3), farm population (4–6), and total population (7). Columns vary the MA exclusion threshold ($J = 3, 5, 10$). Controls as in Table 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; SE clustered by county.

Figure 7: Marginal Effect of Market Access on Population by IP Proximity



Notes: Marginal effect $\hat{\beta}_1 + \hat{\beta}_2 \cdot \ln \text{IP dist}$ from equation (4) with $J = 5$, evaluated over the empirical distribution of log IP distance. Dotted vertical lines mark quartiles. Inner band = 90% CI; outer = 95% CI. Delta-method standard errors.

Age cohort dynamics. To identify *who* responds to market access changes, Table 8 estimates equation (2) separately for eight birth cohorts defined by their 1970 age group, using the log change in cohort size between 1970 and 1990 as the outcome. A positive coefficient indicates net retention or immigration; a negative coefficient indicates outmigration. Coefficients are near zero and insignificant for the youngest cohorts (≤ 19 in 1970), whose dynamics are dominated by demographics. For prime working-age cohorts (25–49 in 1970), the MA coefficient is positive, significant, and stable at roughly 0.33–0.38 log points, indicating that townships with greater road-driven connectivity retained their working-age agricultural population. These are the cohorts who were active farmers in 1970 and middle-aged by 1990 — precisely those who would benefit most from rising agricultural productivity and mechanisation.

Table 8: Market Access and Age Cohort Retention, 1970–1990

Table 9:

	Age 15 (1970 def.)	Age 15–19	Age 20–24	Age 25–29	Age 30–34	Age 35–39	Age 40–44	Age 45–49
log(D MA excl. 10)	0.159 (0.176)	0.150 (0.163)	0.266* (0.141)	0.316*** (0.111)	0.384*** (0.109)	0.384*** (0.107)	0.329*** (0.107)	0.329*** (0.104)
Observations	1307	1307	1307	1307	1307	1307	1307	1307
R ²	0.065	0.075	0.073	0.052	0.043	0.039	0.040	0.044

Notes: OLS estimates of equation (2) with $\Delta \log(\widetilde{\text{MA}}_i^{10})$. Dep. var.: log change in the size of each birth cohort (defined by 1970 age group) between 1970 and 1990. A positive coefficient indicates population retention or net in-migration. Controls as in Table 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; SE clustered by county.

3.4 Service Sector Growth

Table 10 examines the change in registered service establishments between 1974 and 1987. A one-unit increase in log MA growth generates approximately 8–9 additional new service businesses per township (preferred: col. 2; s.e. 3.8, $t = 2.33$), economically large relative to a median of six new establishments. This result is consistent with the migration findings: in townships where MA improvements support rural population retention, higher incomes from mechanised agriculture generate local demand for non-traded services.

Table 10: Service Sector Growth, 1974–1987

Table 11:

	(1) N services	(2) N services	(3) N services
log(D MA excl. 5) [preferred]	3.442 (3.729)	8.779** (3.766)	
log(D MA excl. 10)			9.812** (4.305)
Observations	1307	1307	1307
R ²	0.271	0.303	0.302

Notes: OLS estimates of equation (2). Dep. var.: change in number of registered service businesses, 1974–1987. Col. (1): geography controls only. Cols. (2)–(3) add log IP distance and vary J .

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; SE clustered by county.

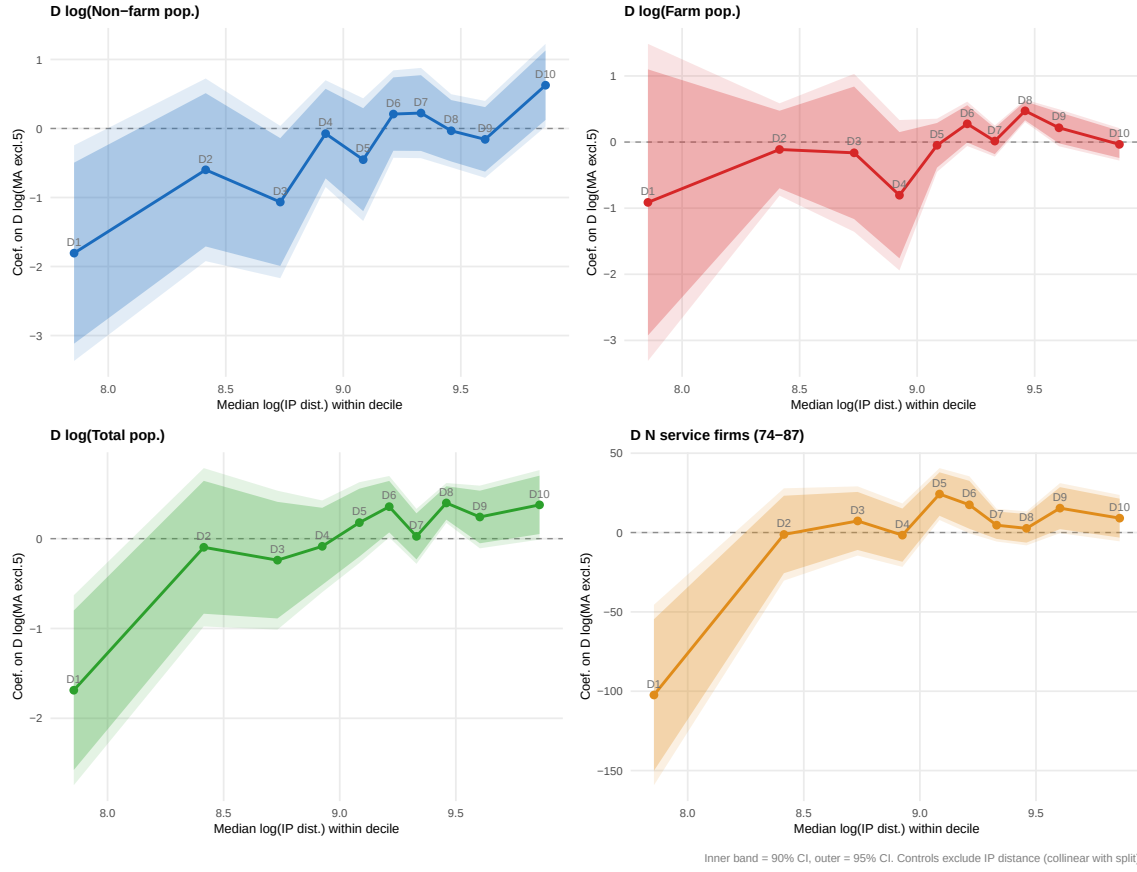
4 Heterogeneity by Industrial Park Proximity

The interaction estimates and the marginal effects in Figure 7 suggest that the role of market access depends critically on a township’s position relative to industrial parks. To characterise this heterogeneity without functional-form assumptions, Figure 8 plots OLS estimates of equation (2) estimated separately within each decile of the log IP-distance distribution (D1 = closest, D10 = farthest).

Figure 8: MA Effect on Population Migration and Service Growth by IP Distance Decile

Population Migration and Service Growth by IP Distance Decile

D1 = closest to industrial park | D10 = farthest | MA = $\log(D \text{ MA, pop.} - \text{fixed, excl. 5 cities})$ | OLS | Controls: geography | SE clustered by county



Notes: OLS coefficient on $\Delta \log(\widetilde{\text{MA}}_i)$ from equation (2) estimated separately within each decile of log IP distance (D1 = nearest IP, D10 = farthest). The x-axis reports the median log IP distance within each decile; decile labels are shown above each point. Log IP distance is excluded from the control set (collinear with the decile partition). Inner band = 90% CI; outer = 95% CI; SE clustered by county.

The non-farm gradient is striking. The MA coefficient falls from -1.81 in D1 (s.e. 0.80 , $t = -2.27$) to $+0.63$ in D10 (s.e. 0.31 , $t = 2.05$), with a sign reversal around the median of the IP-distance distribution. Near industrial parks, road improvements primarily facilitate worker reallocation into manufacturing. In remote townships, the same connectivity improvements support local non-farm activity. The service sector panel reinforces this interpretation: the MA effect on new service establishments is positive in middle-to-remote deciles (D5–D10) and essentially flat near industrial parks (D1–D3), consistent with rising rural incomes and population retention — rather than manufacturing demand — driving local service creation. Farm population follows a similar middle-distance concentration (strongest retention at D6–D8), where farming remains the primary livelihood and output market access genuinely raises its returns.

Machinery adoption (not shown) follows an inverted-U centred on D5–D8, consistent with mechanisation being most attractive where road connectivity improves and agricultural incentives are preserved.

5 Robustness

If our results reflected pre-existing geographic advantages rather than road-driven connectivity improvements, $\Delta \log(\widetilde{\text{MA}}_i^J)$ should predict the *1970 stock* of farm machinery — prior to any road expansion. Table 12 tests this placebo. Across all three MA specifications, the coefficient on the pre-treatment tiller count is small and statistically insignificant (all $|t| < 1.7$), while column (4) shows the main result for comparison. The absence of pre-trend supports the causal interpretation of the main estimates. Coefficient stability across $J \in \{3, 5, 8, 10\}$ throughout Tables 2–10 provides additional reassurance that local road endogeneity does not drive the findings.

Table 12: Pre-Treatment Placebo: 1970 Tiller Stock vs. Market Access Change

Table 13:

	Placebo: HH tillers 1970 (all MA)	Placebo: HH tillers 1970 (excl. 3)	Placebo: HH tillers 1970 (excl. 5)	Main: D HH tillers 70-90 (excl. 5)
log(D(74-87) MA)	0.155			
	(0.310)			
log(D MA excl. 3)		0.041		
		(0.326)		
log(D MA excl. 5) [preferred]			0.035	271.607***
			(0.342)	(68.271)
log(water source area, 1970)				79.016***
				(11.523)
log(population, 1970)				111.533***
				(22.444)
log(effective road)				-39.585***
				(10.391)
Paved road in 30km buffer				0.000
				(0.000)
Observations	1307	1307	1307	1307
R ²	0.083	0.083	0.083	0.457

Notes: Dep. var. in cols. (1)–(3): number of households owning tillers in 1970 (pre-treatment stock). Col. (4): main result — change in households owning tillers, 1970–1990. Significant coefficients in cols. (1)–(3) would indicate pre-existing sorting bias. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; SE clustered by county.